Proposal-first PIDray-Occlusion-aware Bilayer SAM

This is the concise equation guide for the proposed architecture.

Open the Draw.io architecture

1. Provenance

Label	Meaning
PIDray	Taken from the PIDray paper or released code
Occ	Taken from the Occlusion-aware paper or code
Proposed	New connection or rule introduced by this hybrid

2. Symbols

Symbol	Meaning
i	image index
j	proposal or object index
$k \in \{1,2,3\}$	Cascade stage
$\ell \in \{1,\dots,5\}$	AttFPN level
x_i	input X-ray image
$F_i^{1:5}$	PIDray AttFPN features
ϕ_i	image-level prohibited confidence
$\hat{c}_{ij}, \hat{b}_{ij}$	predicted class and box
s_{ij}^{det}	detection confidence
f_i^{img}	SAM image embedding
f_{ij}^{sparse}	learned sparse prompt
$\hat{m}_{ij}^{e}$	overlap or occluder mask
$\hat{m}_{ij}$	final target mask

The two feature systems must not be confused:

$$\boxed{F_i^{1:5} \neq f_i^{img}}$$

3. Training annotations

For image x_i:

$$Y_i = \left\{(b_{ij}, c_{ij}, m_{ij}, m_{ij}^e)\right\}_{j=1}^{n_i}, \qquad y_i^0 = \mathbf{1}[n_i > 0]$$

- m_{ij} is the target-object mask.
- m_{ij}^e is the overlapping part of another annotated prohibited object.
- y_i^0 is the binary label used by PIDray R_0.

4. PIDray backbone and AttFPN

$$\{C_i^2, C_i^3, C_i^4, C_i^5\} = \Phi_{\text{ResNet101}}(x_i)$$

$$\boxed{\{F_i^1, \ldots, F_i^5\} = \Phi_{\text{AttFPN}}(C_i^2, C_i^3, C_i^4, C_i^5)}$$

Each F_i^ℓ has 256 channels. A compact description of AttFPN is:

$$\begin{aligned}
G_{i,\ell r} &= \text{Resize}(\widetilde{F}_i^r, H_\ell, W_\ell), \\
C_i^\ell &= \sum_{r=1}^{5} A_{i,\ell r}^{ch} \odot G_{i,\ell r}, \\
S_i^\ell &= \sum_{r=1}^{5} A_{i,\ell r}^{sp} \odot G_{i,\ell r}, \\
F_i^\ell &= \widetilde{F}_i^\ell + \text{DR}_\ell(C_i^\ell + S_i^\ell).
\end{aligned}$$

This compact equation summarizes the implementation. PIDray describes the operations separately as channel attention, spatial attention, and dependency refinement.

5. PIDray coarse node R_0

Paper formulation

$$F_i^0 = \text{Agg}(F_i^1, \ldots, F_i^5), \qquad E_{ip} = F_{ip}^0 + PE_p$$

For attention head r:

$$\begin{aligned}
\alpha_{ip}^{(r)} &= \text{softmax}_p\left(\frac{\langle W_Q^{(r)} Q, W_K^{(r)} E_{ip}\rangle}{\sqrt{d}}\right), \\
h_i^{(r)} &= \sum_{p=1}^{HW} \alpha_{ip}^{(r)} W_V^{(r)} F_{ip}^0, \\
V_i &= \text{FFN}\left[W_H\, \text{Cat}(h_i^{(1)}, \ldots, h_i^{(N_h)})\right].
\end{aligned}$$

The final probability and decision are:

$$\boxed{\phi_i = \sigma(w_\phi^\top V_i + b_\phi)}, \qquad g_i^0 = \mathbf{1}[\phi_i > 0.5]$$

Released-code difference

The released PIDray branch uses:

$$\boxed{F_i^0 \leftarrow F_i^5}$$

Its inference path runs RPN and RoI processing before suppressing detections when $\phi_i \le 0.5$. Therefore, released R_0 is not a detector compute-saving early exit.

6. RPN and three-stage Cascade

The RPN produces:

$$\boxed{\hat{\mathcal{O}}_i = \Phi_{\text{rpn}}(F_i^{1:5}) = \{\hat{o}_{ij}\}_{j=1}^{M_i}}$$

Initialize $\hat{b}_{ij}^{(0)} = \hat{o}_{ij}$. For Cascade stage k:

$$\begin{aligned} u_{ij}^{(k)} &= \Phi_{\text{roi}}^{(k)}(F_i^{1:4}, \hat{b}_{ij}^{(k-1)}), \\ z_{ij}^{(k)} &= \Phi_{\text{cls}}^{(k)}(u_{ij}^{(k)}), \\ \Delta b_{ij}^{(k)} &= \Phi_{\text{reg}}^{(k)}(u_{ij}^{(k)}), \\ \hat{b}_{ij}^{(k)} &= \text{Decode}(\hat{b}_{ij}^{(k-1)}, \Delta b_{ij}^{(k)}). \end{aligned}$$

The stages use IoU thresholds 0.5, 0.6, and 0.7. The final detector returns:

$$\boxed{(\hat{c}_{ij}, s_{ij}^{det}, \hat{b}_{ij}), \qquad \hat{b}_{ij} = \hat{b}_{ij}^{(3)}}$$

7. Proposed proposal screening

This gate is new:

$$q_{ij} = \phi_i s_{ij}^{det}$$

$$\boxed{g_{ij} = \mathbf{1}[\phi_i > \tau_0]\mathbf{1}[s_{ij}^{det} > \tau_d], \qquad \mathcal{K}_i = \{j : g_{ij} = 1\}}$$

- ϕ_i provides global image evidence.
- s_{ij}^{det} provides local proposal evidence.
- q_{ij} ranks retained proposals.
- If $\mathcal{K}_i = \varnothing$, inference skips the SAM/Occ branch.

8. Detector-to-SAM bridge

For each retained proposal $j \in \mathcal{K}_i$:

$$\boxed{\hat{v}_{ij} = \Phi_{\text{roi}}^{mask}(F_i^{1:4}, \hat{b}_{ij})}$$

$$\boxed{f_{ij}^{sparse} = \Phi_{\text{prompt}}(\hat{v}_{ij}), \qquad f_{ij}^{sparse} \in \mathbb{R}^{5\times 256}}$$

The local RoI feature is $256 \times 14 \times 14$.

9. SAM encoder

The image embedding is computed once and reused:

$$\boxed{f_i^{img} = \Phi_{\text{enc}}(\mathcal{P}_S(x_i))}$$

For the local ViT-L model with a 512×512 input:

$$f_i^{img} \in \mathbb{R}^{256 \times 32 \times 32}$$

The original Occ pipeline uses f_i^{img} for RPN and RoI processing. The hybrid instead uses PIDray $F_i^{1:5}$ for detection and keeps f_i^{img} for mask decoding.

10. Occlusion-aware bilayer decoding

Occluder decoder

$$T_{ij}^r = \text{Cat}(t_{\text{mask}}, t_{\text{IoU}}, f_{ij}^{sparse})$$

$$\boxed{(\hat{m}_{ij}^e, f_{r,ij}^{img}, f_{r,ij}^{sparse}) = \Phi_r(T_{ij}^r, f_i^{img})}$$

Residual guidance

$$\boxed{f_{e,ij}^{img} = f_i^{img} + f_{r,ij}^{img}, \qquad f_{e,ij}^{sparse} = f_{ij}^{sparse} + f_{r,ij}^{sparse}}$$

The overlap mask is an auxiliary output, not a direct input to Φ_e:

$$\boxed{\hat{m}_{ij}^e \not\to \Phi_e}$$

Occludee decoder

$$T_{ij}^e = \text{Cat}(t_{\text{mask}}, t_{\text{IoU}}, f_{e,ij}^{sparse})$$

$$\boxed{\hat{m}_{ij} = \Phi_e(T_{ij}^e, f_{e,ij}^{img})}$$

Token-order note

The Occ paper writes:

$$\text{Cat}(t_{\text{mask}}, t_{\text{IoU}}, f^{sparse})$$

The local decoder packs:

$$[t_{\text{IoU}}, t_{\text{mask}}^{1:4}, f_{ij}^{sparse}]$$

This gives $1 + 4 + 5 = 10$ tokens. The paper and code describe the same token roles but list them differently.

11. Final output

Restore the target mask to the original image:

$$\hat{m}_{ij}^{orig} = T_i^{-1}\left[\text{Upsample}(\hat{m}_{ij})\right]$$

$$\boxed{\hat{\mathcal{Y}}_i = \left\{ (\hat{c}_{ij}, \hat{b}_{ij}, \hat{m}_{ij}^{orig}, \hat{m}_{ij}^{e}, q_{ij}) : j \in \mathcal{K}_i \right\}}$$

The primary outputs are class, box, and target mask. The overlap mask is mainly used for training and diagnosis.

12. Training loss

$$\mathcal{L}_{R_0} = \mathrm{BCE}(\phi_i, y_i^0)$$

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{rpn}} + \sum_{k=1}^{3} w_k \left[\mathcal{L}_{\mathrm{cls}}^{(k)} + \mathbf{1}_{ij}^{+} \mathcal{L}_{\mathrm{reg}}^{(k)} \right]$$

where $(w_1, w_2, w_3) = (1, 0.5, 0.25)$.

$$\mathcal{L}_{\mathrm{bilayer}} = \frac{1}{N_+} \sum_{(i,j) \in \mathcal{P}^+} \left[\mathcal{L}_{\mathrm{pred}}(\hat{m}_{ij}, m_{ij}) + \lambda_e \mathcal{L}_{\mathrm{pred}}(\hat{m}_{ij}^{e}, m_{ij}^{e}) \right]$$

The local Occ code uses $\lambda_e = 0.25$. A starting hybrid objective is:

$$\boxed{\mathcal{L}_{\mathrm{total}} = 0.1 \mathcal{L}_{R_0} + \mathcal{L}_{\mathrm{det}} + \mathcal{L}_{\mathrm{bilayer}}}$$

This combined loss is proposed; it is not printed by either paper.

13. Training versus inference

Training uses matched positive RoIs:

$$g_{ij}^{train} = \mathbf{1}[(i,j) \text{ is a matched positive RoI}]$$

Do not use the predicted hard R_0 gate to remove genuine positive mask examples.

Inference uses:

$$g_{ij}^{test} = \mathbf{1}[\phi_i > \tau_0] \mathbf{1}[s_{ij}^{det} > \tau_d]$$

14. Component ownership

Component	Source
ResNet-101, AttFPN, R_0, RPN, Cascade	PIDray
SAM encoder, prompt head, Φ_r, residual guidance, Φ_e	Occ/RSPrompter
Detector-to-SAM bridge and proposal screening	proposed hybrid
Skipping SAM when no proposal survives	proposed hybrid
Combined total loss	proposed hybrid

The original PIDray FCN mask head is excluded because Φ_e becomes the final mask head.

Training $\hat{m}_{ij}^{e}$ requires PIDray-A-style overlap annotations. Standard PIDray masks alone cannot supervise this auxiliary loss.

15. Complete flow

$$\boxed{x_i \xrightarrow{\text{ResNet101+AttFPN}} F_i^{1:5} \xrightarrow{R_0+\text{RPN+Cascade}} (\phi_i, \hat{c}_{ij}, s_{ij}^{det}, \hat{b}_{ij}, \hat{v}_{ij}) \xrightarrow{\text{screening}+\Phi_{\text{prompt}}} f_{ij}^{sparse}}$$

$$\boxed{x_i \xrightarrow{\Phi_{\text{enc}}} f_i^{img}, \qquad (f_i^{img}, f_{ij}^{sparse}) \xrightarrow{\Phi_r \to \text{residual} \to \Phi_e} (\hat{m}_{ij}^{e}, \hat{m}_{ij})}$$

The PIDray branch determines **where and what**. The bilayer SAM branch predicts the target mask using overlap-aware guidance.

References

- PIDray paper
- Occlusion-aware paper
- PIDray configuration
- PIDray R_0 implementation
- AttFPN implementation
- Occ configuration
- Occ bilayer decoder